

A universal fractional cover envelope and its design equality cases

Research continuation, 7 October 2026. All inequalities in this note are proved for fractional covers. No small-intersection hypothesis is used, and no corresponding integer-cover bound is asserted.

1. Definitions and finite theorem

Let H be a finite simple intersecting r -uniform hypergraph, $r \geq 2$, with $m \geq 1$ edges. Ignore vertices outside its edges. A fractional vertex cover assigns nonnegative weights z_v to vertices so that the sum on each edge is at least one; its minimum total is $\tau^*(H)$. Its finite linear-programming dual assigns nonnegative weights y_E to edges, with sum at each vertex at most one, and maximizes $Y = \sum_E y_E$. Appendix A explains the duality used here.

Put $k = \lceil m/r \rceil$. If $k = 1$, pairing gives $\tau^*(H) \leq \tau(H) \leq \lceil m/2 \rceil \leq m/2 + 1/2$. For $k \geq 2$ define

$$g = (k-1)r + 1, \quad U_k(r, m) = \frac{k(k-1)r^2 + (2k-1)r + 1 - m}{(k^2 + k - 1)r + 1 - m}. \quad (1)$$

The choice of k gives $g \leq m \leq kr$, and the denominator is positive.

Theorem 1 (universal finite envelope and equality). For $k \geq 2$,

$$\tau^*(H) \leq U_k(r, m). \quad (2)$$

Equality holds if and only if $m = g$, H is linear, and every active vertex has degree k . Thus equality hypergraphs are precisely the incidence duals of Steiner 2- $(m, k, 1)$ designs with replication r . No resolvability is required for this fractional equality statement. If $m > g$, inequality (2) is strict.

Define a continuous function ϕ on nonnegative real numbers by

$$\phi(c) = c/2 \quad (0 \leq c \leq 1), \quad \phi(c) = \frac{k(k-1)}{k^2 + k - 1 - c} \quad (k-1 \leq c \leq k, \quad k \geq 2). \quad (3)$$

The definitions agree at shared endpoints. In particular $\phi(a) = a/(a+1)$ for positive integers a .

Corollary 2 (bounded edge/rank ratios). For $k \geq 2$,

$$\tau^*(H) \leq r\phi(m/r) + 1/k. \quad (4)$$

For any sequence with r tending to infinity and m/r tending to a finite $c \geq 0$,

$$\limsup \tau^*(H)/r \leq \phi(c). \quad (5)$$

This has no partite, linearity or intersection-excess assumption.

Let h be the piecewise linear interpolation of $a/(a+1)$ at nonnegative integers a . For $k-1 < c < k$, $k \geq 2$, put $\theta = c - k + 1$. Then

$$h(c) - \phi(c) = \frac{\theta(1 - \theta)}{k(k+1)(k^2 - \theta)} > 0. \quad (6)$$

Thus (5) is strictly smaller than both $h(c)$ and $c/(1+c)$ at noninteger $c > 1$. Neither ϕ nor the strict finite bound for $m > g$ is claimed to be the optimal envelope at these ratios. In fact the universal strict gap in Theorem 3 below rules out attaining ϕ there.

2. Two weighted counting observations

Fix any feasible dual weights, set $Y = \sum_E y_E$, and let $b = \max_E y_E$. Choose an edge P with $y_P = b$. Since each edge other than P meets P and weights are nonnegative,

$$Y - b \leq \sum_{v \in P} \sum_{E \neq P: v \in E} y_E \leq r(1 - b), \quad Y \leq r - (r - 1)b. \quad (7)$$

The first inequality remains valid with multiple intersections. No equality or linearity is silently assumed.

Suppose $b > 1/(k+1)$ and put $t = (1 - b)/k$, so $b > t$. Let G be the set of edges with weight strictly greater than t ; P belongs to G . At a vertex of P there can be at most $k-1$ other edges of G : k of them would have total weight greater than $kt = 1 - b$, violating feasibility. All other members of G meet P . Consequently

$$|G| \leq 1 + r(k - 1) = g. \quad (8)$$

Every weight in G is at most b , and every weight outside G is at most t . Because $b > t$ and $g \leq m$, (8) gives

$$Y \leq |G|b + (m - |G|)t \leq gb + (m - g)(1 - b)/k. \quad (9)$$

These statements require only finite nonnegative sums and vertex constraints. In particular they hold for every feasible dual vector, not just a solver's reported optimum.

3. Derivation of the finite envelope

First suppose $b \leq 1/(k+1)$. Then $Y \leq m/(k+1)$. The numerator N and denominator Q in (1) satisfy

$$(k+1)N - mQ = (m - (k+1)g)(m - kr - 1) > 0, \quad (10)$$

because $g = (k-1)r + 1$, $r \geq 2$, $k \geq 2$ and $m \leq kr < (k+1)g$. Therefore $Y < U_k(r, m)$ in this case.

Otherwise use (9). The coefficient

$$A = ((k+1)g - m)/k$$

of b is positive. By (7), $b \leq (r - Y)/(r - 1)$. Substituting in (9) and multiplying positive denominators gives

$$Y\{k(r - 1) + (k+1)g - m\} \leq g(kr + 1) - m. \quad (11)$$

The braces equal Q and the right side equals N . This proves $Y \leq U$. Theorem 1's inequality now follows by finite LP duality. Its proof does not use a numerical solver, a colouring theorem or a literature claim about another research result.

4. Complete finite equality analysis

Suppose $\tau^* = U$. An optimum dual vector exists: each coordinate lies in $[0,1]$, so its feasible set is a nonempty compact polytope. By (10) its maximum b is greater than $1/(k+1)$. Equality in (11) forces equality in (7) and (9), since $A > 0$. In particular

$$|G| = g, \quad y_E = b \ (E \in G), \quad y_E = t = (1 - b)/k \ (E \notin G). \quad (12)$$

Every member of G other than P meets P . Equality in (8) forces each of P 's r vertices to belong to exactly $k-1$ other G edges, with each G edge meeting P exactly once. Equality in the last inequality of (7) forces the total incident dual weight at every vertex of P to be one.

But a vertex of P already lies in k edges of G . If it also lay in one outside edge, its total weight would be at least

$$kb + t = kb + (1 - b)/k > 1, \quad (13)$$

where the strict inequality is equivalent to $b > 1/(k + 1)$. Thus no outside edge can meet P . Intersectingness implies that there are no outside edges, so $m = g$. Equation (12) gives uniform edge weights b , and the saturation at P gives $kb = 1$. Thus $b = 1/k$ and $Y = m/k$.

Apply (7) to each original edge, all of which now have maximum weight b . Since $Y - b = (m - 1)/k = r(k - 1)/k = r(1 - b)$, every inequality in (7) is an equality. Positive edge weights force every pair of distinct original edges to intersect exactly once. Saturation at every active vertex gives $d(v)b = 1$, hence $d(v) = k$. This proves the necessary equality conditions.

Conversely, if H is linear, all active degrees are k , and $m = g$, uniform dual weights $1/k$ are feasible with total m/k . Uniform primal vertex weights $1/r$ give every edge total one; double counting incidences gives $n = mr/k$, so their total $n/r = m/k$. Weak duality certifies $\tau^* = m/k$. At $m = g$, (1) equals m/k . This proves the converse without requiring design existence.

The design interpretation is explicit: original edges become points, and original vertices become k -point blocks. Linearity and intersectingness say every pair of points belongs to exactly one block. Each point belongs to exactly r blocks. Conversely the incidence dual of such a design has all the stated properties.

5. Asymptotic comparison and examples

Write $c = m/r = k - 1 + \theta$ with $0 < \theta \leq 1$ and $k \geq 2$. Set $d = k^{2-\theta}$ and $a = k(k-1)$, so $\phi(c) = a/d$. Subtracting $r\phi(c)$ from (1) gives

$$U_k - r\phi(c) = \frac{(k - \theta - \phi(c))r + 1}{dr + 1}. \quad (14)$$

This is at most $1/k$: after multiplication, the difference between $(dr+1)/k$ and the numerator is

$$r\{\theta(1 - 1/k) + \phi(c)\} - (k - 1)/k \geq 0, \quad (15)$$

since $\phi(c) \geq (k - 1)/k$ and $r \geq 1$. This proves (4). At $c \leq 1$ use pairing. The continuity in (3) then proves (5), including $c=0$. The identity (6) follows by expanding $(k^{2-1+\theta})(k^{2-\theta}) - k^{2(k-1-\theta)} = \theta(1-\theta)$.

For a prime power s , the affine-line design on F_s^N has $m = s^N$ points, block size s and replication $r = (s^N - 1)/(s - 1)$. Its dual is linear intersecting r -uniform with degree s and $\tau^* = m/s$. Here $k = s$ and

$m=(s-1)r+1$. A parallel class also certifies the integer cover $\tau = m/s$, but resolvability was not used above. These are standard design examples, not new constructions.

The projective plane of order q has $r=q+1$, $m=q^2+q+1$, $k=r$ and $m=(k-1)r+1$; it also attains (2). In particular the Fano plane gives $r=3, m=7, \tau^* = 7/3$. This finite boundary is useful: one cannot drop the finite correction and claim $\tau^* \leq \phi(m/r)$ for every finite hypergraph.

6. A uniform explicit gap at noninteger ratios

Theorem 3. Fix $k-1 < c < k$, $k \geq 2$. Every sequence in Corollary 2 has $\limsup \tau^*/r \leq \phi(c) - \zeta_c$ for the positive constant below. No small-intersection hypothesis is added.

Set $\theta = c - k + 1$, $p = \phi(c)$, and define

$$a_0 = k - 1 - \theta/k, \quad b_0 = 1 - p, \quad t_0 = p/k, \quad d_0 = b_0 - t_0 > 0,$$

$$e_0 = \min\{t_0/4, d_0/4, (k-1)d_0/[4(k+1)]\}, \quad \delta_0 = \min\{a_0 d_0/[8(k+1)], kt_0/4\}, \quad M = \lceil 2/t_0 \rceil.$$

Then a conservative explicit choice is

$$\zeta_c = \min\left\{\delta_0, \frac{\theta}{(a_0 + 1)\{2M/d_0 + (M + 1)/e_0\}}\right\} > 0. \quad (16)$$

Proof. Pass to a subsequence on which τ^*/r tends to its upper limit α , choose optimal dual weights, and pass further so their maxima b_r tend to b . The ceiling of m/r is eventually the fixed k . Set $\delta = p - \alpha \geq 0$ by Corollary 2. If $\delta \geq \delta_0$ the conclusion already holds, so assume $\delta < \delta_0$.

The small-maximum branch gives $\alpha \leq c/(k+1)$ if $b \leq 1/(k+1)$. But

$$p - c/(k+1) = a_0 k d_0 / (k+1) > \delta_0.$$

Thus $b > 1/(k+1)$. For all sufficiently large indices use the threshold $t_r = (1 - b_r)/k$, the set G from Section 2, and $g = (k-1)r + 1$. Define the nonnegative mass defect

$$D_r = g b_r + (m - g) t_r - Y = (g - |G|)(b_r - t_r) + \sum_{E \in G} (b_r - y_E) + \sum_{E \notin G} (t_r - y_E). \quad (17)$$

The star bound and (9), after taking limits, give

$$b_0 - \delta/a_0 \leq b \leq b_0 + \delta, \quad \lim D_r/r = a_0 b + \theta/k - \alpha \leq (a_0 + 1)\delta. \quad (18)$$

Here $a_0 b_0 + \theta/k = p$. All constants are fixed before taking limits; no uniform asymptotic input is needed.

Call a G edge high-good if $y_E \geq b_r - e_0$, and an edge outside G low-good if $y_E \geq t_r - e_0$. Let their numbers be H_r and L_r . Equation (17) yields

$$H_r \geq g - \frac{D_r}{b_r - t_r} - \frac{D_r}{e_0}, \quad L_r \geq m - g - \frac{D_r}{e_0}. \quad (19)$$

The chosen δ_0 and (18) imply $b_r - t_r \geq d_0/2$ eventually, and $t_r - e_0 > t_0/2$. They also imply

$$(k+1)(b_r - e_0) > 1, \quad k(b_r - e_0) + (t_r - e_0) > 1. \quad (20)$$

For explicit verification, the unperturbed margins in (20) are kd_0 and $(k-1)d_0$. The possible lower change in b is at most $\delta/a_0 < d_0/[8(k+1)]$, and $(k+1)e_0 \leq (k-1)d_0/4$. The asserted strict margins follow. The lower bound on t_r-e_0 uses $\delta/k < t_0/4$ and $e_0 \leq t_0/4$.

Choose an edge P of maximum weight. It is high-good. Every vertex of P meets at most $k-1$ other high-good edges by the first inequality in (20). A vertex meeting a low-good edge meets at most $k-2$ other high-good edges by the second. Each such vertex meets at most M low-good edges because each has weight greater than $t_0/2$. All high-good and low-good edges meet P , so

$$H_r - 1 \leq (k-1)r - L_r/M. \quad (21)$$

Repeated intersections can only increase the incident counts used in this inequality and do not invalidate it.

Insert (19) into (21), divide by r , and use (18) and $b_r - t_r \geq d_0/2$. This gives

$$\theta \leq (a_0 + 1)\delta\{2M/d_0 + (M+1)/e_0\}.$$

Together with the earlier $\delta \geq \delta_0$ case this proves (16). Every subsequential upper limit satisfies the result, completing the proof. The displayed gap is not claimed optimal.

7. Implications and limitations

For m/r tending to $c>0$ and integer cover τ/r tending to one, (5) forces

$$\liminf \tau/\tau^* \geq 1/\phi(c). \quad (22)$$

The denominator is positive since $\tau \geq 1$ implies $\tau^* \geq 1$, and the asymptotic statement follows by division. This describes a necessary integrality gap for sparse maximal-cover families, including possible partite Ryser counterexamples. It does not exclude such families.

Kahn's 1994 Theorem 5.2 rounds a fractional cover of a fixed-rank auxiliary hypergraph only when its *weighted* pair codegrees tend to zero. An arbitrary optimal fractional cover need not satisfy that condition. Thus (5) cannot simply be advertised as an integer-cover theorem, even under small unweighted intersections. The earlier near-linear integer envelope in this repository has a separate proof and a separate hypothesis $I = o(r^2)$.

The proof above is self-contained apart from the standard finite convex separation used to explain LP duality in Appendix A. Exact rational primal/dual certificates and partial Lean algebra exports accompany the note. They do not replace the universal counting proof. Historical novelty and the optimal noninteger frontier remain under investigation; no priority is asserted.

Appendix A. Finite LP duality used here

Let A be the vertex-by-edge incidence matrix. Weak duality follows from $y^\top A^\top z \leq \mathbf{1}^\top z$ and $y^\top A^\top z \geq \mathbf{1}^\top y$ for feasible z, y . The primal infimum α is attained: clipping each z_v to at most one preserves all constraints and leaves a compact feasible cube.

For completeness consider the finitely generated closed convex cone

$$C = \{(A^\top z - s, \mathbf{1}^\top z + t) : z, s, t \geq 0\} \subset \mathbb{R}^{m+1}.$$

For $\beta < \alpha$ the point $(\mathbf{1}, \beta)$ is outside C . Finite-dimensional separation supplies (a, b) nonnegative on C and strictly negative on that point. The generators $(-e_i, 0)$ and $(0, 1)$ force $a_i \leq 0$ and $b \geq 0$.

Write $y = -a$. The generators associated with each vertex force $Ay \leq b\mathbf{1}$, and strict separation gives $\sum_i y_i > b\beta$. If $b = 0$, then $Ay \leq 0$ and $y \geq 0$; every edge is nonempty, so $y = 0$, a contradiction. Hence y/b is dual-feasible with value $>\beta$. Letting β increase to α and using compactness of the dual feasible cube proves equality of the optima.

The standard facts just used are finite convex separation and the closedness of a finitely generated cone. For the latter, any point of such a cone has a representation using linearly independent generators, by deleting dependencies while preserving nonnegative coefficients. Along a convergent sequence pass to one fixed independent subset (there are finitely many); the coefficients converge by continuity of the inverse on its span, proving closedness. No asymptotic combinatorial theorem enters this step.

References and provenance

[K] J. Kahn, *On a Problem of Erdős and Lovász. II: $n(r)=O(r)$* , J. Amer. Math. Soc. **7** (1994), 125–143, DOI <https://doi.org/10.1090/S0894-0347-1994-1224593-5>, Section 5. Its harmonic integer-cover bound and weighted rounding are prior results; neither is used to prove Theorem 1 here.

[FKS] Z. Füredi, J. Kahn and P. D. Seymour, *On the fractional matching polytope of a hypergraph*, Combinatorica **13** (1993), 167–180, DOI <https://doi.org/10.1007/BF01303202>. Fractional matching/cover theory is the classical setting; this reference is not used as an unproved input for (2).

[Previous] Repository note *An asymptotic cover law for sparse nearly linear intersecting hypergraphs*, commit 34a212f, with the subsequent Kahn prior-work addendum. Its h bound is for integer covers under a different hypothesis. The present fractional bound has its own proof and complete finite equality analysis.